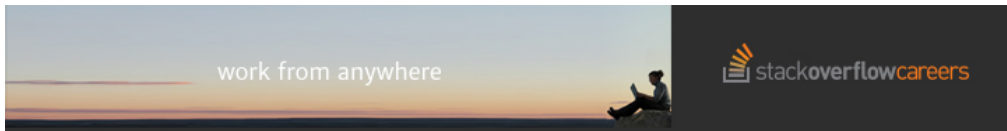


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How to use PYTHONPATH



How can I make any use of PYTHONPATH? When I try to run a script in the path the file is not found. When I cd to the directory holding the script the script runs. So what good is the PYTHONPATH?

```
$ echo $PYTHONPATH
:/home/andy/lib/python

$ tree -L 1 '/home/andy/lib/python'
/home/andy/lib/python
├── gbm_html.py
├── gbm.py
├── __init__.py
├── __pycache__
├── scripts
└── yesno.py

$ python gbm.py -h
python: can't open file 'gbm.py': [Errno 2] No such file or directory

$ cd '/home/andy/lib/python'
```

After cd to the file directory it runs ..

```
$ python gbm.py -h
usage: gbm.py [-h] [-b]
```

Why can I not make any use of the PYTHONPATH?

python unix

edited Nov 11 '13 at 22:55



Southpaw Hare

768 1 8 30

asked Nov 11 '13 at 22:45



Randy Skretka

1,005 2 9 13

3 Answers

I think you're a little confused. PYTHONPATH sets the search path for **importing** python modules, not for executing them like you're trying.

PYTHONPATH Augment the default search path for module files. The format is the same as the shell's PATH: one or more directory pathnames separated by os.pathsep (e.g. colons on Unix or semicolons on Windows). Non-existent directories are silently ignored.

In addition to normal directories, individual PYTHONPATH entries may refer to zipfiles containing pure Python modules (in either source or compiled form). Extension modules cannot be imported from zipfiles.

The default search path is installation dependent, but generally begins with prefix/lib/pythonversion (see PYTHONHOME above). It is always appended to PYTHONPATH.

An additional directory will be inserted in the search path in front of PYTHONPATH as described above under Interface options. The search path can be manipulated from within a Python program as the variable sys.path.

<http://docs.python.org/2/using/cmdline.html#envvar-PYTHONPATH>

What you're looking for is PATH.

```
export PATH=$PATH:/home/andy/lib/python
```

However, to run your python script as a program, you also need to set a [shebang](#) for Python in the first line. Something like this should work:

```
#!/usr/bin/env python
```

And give execution privileges to it:

```
chmod +x /home/randy/lib/python/gbm.py
```

Then you should be able to simply run `gbm.py` from anywhere.

edited Nov 11 '13 at 22:56

answered Nov 11 '13 at 22:51



Pedro Werneck

9,137 1 16 42

Thanks friend. I am still ironing this out. My script does succeed in importing my modules from my custom PYTHONPATH. And now I understand the difference: it is all about script vs. module; command script call vs. python import. Yes, I will in fact use `export PATH=$PATH:/home/etc` – [Randy Skretka](#) Nov 11 '13 at 23:35

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PYTHONPATH only affects `import` statements, not the top-level Python interpreter's lookup of python files given as arguments.

Needing PYTHONPATH to be set is not a great idea - as with anything dependent on environment variables, replicating things consistently across different machines gets tricky. Better is to use Python 'packages' which can be installed (using 'pip', or distutils) in system-dependent paths which Python already knows about.

Have a read of <http://guide.python-distribute.org/introduction.html> - 'The Hitchhiker's Guide to Packaging', and also <http://docs.python.org/2/tutorial/modules.html> - which explains PYTHONPATH and packages at a lower level.

answered Nov 11 '13 at 22:58



codedstructure

343 2 5

Point taken. In this particular case adding to PYTHONPATH seems to work out nicely though. As one user on one machine I can well make use and reuse of some already written .py code. – [Randy Skretka](#) Nov 11 '13 at 23:51

You're confusing PATH and PYTHONPATH. You need to do this:

```
export PATH=$PATH:/home/randy/lib/python
```

PYTHONPATH is used by the python interpreter to determine which modules to load.

PATH is used by the shell to determine which executables to run.

edited May 14 at 20:03

answered Nov 11 '13 at 22:50



A-B-B

3,772 1 22 23



Richard

1,137 6 19